

MAS · Complex Numbers — Question Patterns and Solution Approaches

Source: marking keys + exam reports for 43 past complex number questions from 2016–2025. Original questions in [topics/01_Complex_Numbers.pdf](#). Numerical examples have been checked with sympy. Also references 2 questions from the 2016 sample paper (2016S-F1, 2016S-A18; original papers in [papers/2016Sample_*](#)). Review status: has undergone one round of independent review and been revised according to feedback (see [Review_log.md](#)).

Marking panel reminders

- **Use polar form when you should** (2016 report: people who did not convert $(\sqrt{3} - i)^5$ to polar form lost a lot of marks). For powers and roots, always use polar form; for multiplication and division, it depends on the form (e.g. for $(2+i)/(1-i)^2$, using the conjugate is faster, 2016F-Q2(a)).
 - **Arguments in the third quadrant** (2020 report specifically names 2020A-Q15(a): complex numbers in the third quadrant converted to polar form incorrectly): e.g. the principal argument of $-1 - i$ is $-3\pi/4$, not $\pi/4$, and it cannot be written as $5\pi/4$ (outside the principal value range). Draw a diagram first to confirm the quadrant.
 - **Vector interpretation of complex numbers** (2018, 2021, 2025 reports: $z + r$, argument of $-z$, regular hexagon): $z + r$, $z - k$ (e.g. 2023A-Q12), and $-z$ must all be done by drawing them as vectors (rhombus, isosceles triangle, regular hexagon).
 - **Basic skill of expanding brackets** (2021, 2023, 2024 reports): $(z - (2+4i))(z - (2-4i)) = z^2 - 4z + 20$ is often calculated incorrectly; $z^2 - 2z + 3 = 0$ must be solved by completing the square or the quadratic formula, and **must not** be split into $(z - 3)(z + 1)$.
 - **Know exact trigonometric values** (2019, 2021, 2023 reports).
 - “Show that $(z - 2i)$ is a factor” : after substituting, you must write out the **calculation of each term**; the key says “provides evidence, i.e. not just the statement $P(2i) = 0$ ”.
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Pattern 1: Algebraic form operations and conversion between forms (mostly in CF)

Approach

- Division: multiply the numerator and denominator by the conjugate of the denominator.
- Converting to polar form: $r = \sqrt{a^2 + b^2}$; θ is determined by the quadrant. Example: given $z = 3 + bi$ and $\tan \theta = 2 \Rightarrow b = 6$, $r = \sqrt{45} = 3\sqrt{5}$ (2024F-Q1).
- Higher powers: $(\sqrt{3} - i)^5 = (2 \operatorname{cis}(-\pi/6))^5 = 32 \operatorname{cis}(-5\pi/6) = -16\sqrt{3} - 16i$. The key also accepts binomial expansion, but it takes several more steps.

Marking points: ✓ modulus ✓ argument ✓ use De Moivre ✓ convert to $a + bi$. **Past questions:** 2016F-Q2, 2017A-Q10(a), 2019A-Q12(a)(b), 2021A-Q11(a)(b), 2024F-Q1.

Pattern 2: Solving $z^n = w$ (finding n th roots) ★ appears every year

Standard method

1. Write w as $R \operatorname{cis} \phi$ (1 mark for modulus, 1 mark for argument).
2. $z = R^{1/n} \operatorname{cis}((\phi + 2k\pi)/n)$, $k = 0, 1, \dots, n-1$ (1 mark for writing the general form).
3. List **all** n roots within the range required by the question (commonly $-\pi < \theta \leq \pi$); the arguments of adjacent roots differ by $2\pi/n$.
4. Example: $z^4 = -16i = 16 \operatorname{cis}(-\pi/2) \Rightarrow z = 2 \operatorname{cis}(-5\pi/8), 2 \operatorname{cis}(-\pi/8), 2 \operatorname{cis}(3\pi/8), 2 \operatorname{cis}(7\pi/8)$. $z^4 = 2 - 2\sqrt{3}i = 4 \operatorname{cis}(-\pi/3) \Rightarrow \sqrt{2} \operatorname{cis}(-7\pi/12), \sqrt{2} \operatorname{cis}(-\pi/12), \sqrt{2} \operatorname{cis}(5\pi/12), \sqrt{2} \operatorname{cis}(11\pi/12)$.

Variations

- Given the diagram of one root, work backwards to find k (2025F-Q1): $w_2^3 = ki$.

- Number of roots in the first quadrant $\Rightarrow$ find the possible values of n , giving reasons (2017A-Q19(b)).
- Roots of $z^n = 1$: on the unit circle, equally spaced, separated by $2\pi/n$ (2024A-Q9(c) has 2 marks for the explanation: circle of radius 1, equal spacing).
- Product of roots (2019F-Q9): product of all roots $= (-1)^{n+1}$; 1 when n is odd, -1 when n is even.
- Common roots: $z^{43} = 1$ and $z^{47} = 1$ have only $z = 1$ as a common root, because 43 and 47 are coprime (2021F-Q7).
- z^n is a positive real number (2022F-Q6(c)): $n \cdot \arg z$ must be an integer multiple of 2π . $z_1 = 4 \operatorname{cis}(5\pi/6) \Rightarrow 5n/6 = 2k \Rightarrow$ smallest $n = 12$, $r = 4^{12} = 2^{24}$.

Past questions: 2016A-Q14, 2017A-Q19, 2018F-Q7(a), 2020A-Q13, 2022A-Q18(b), 2022F-Q6, 2023F-Q6, 2024A-Q9, 2025F-Q1.

Pattern 3: Complex roots of polynomials (factor theorem + conjugate root theorem)

Standard method

1. "Show that $(z - \alpha)$ is a factor" : calculate $P(\alpha)$, **write out each term**, and obtain 0.
2. A polynomial with real coefficients $\Rightarrow$ the conjugate $\bar{\alpha}$ is also a root $\Rightarrow (z - \alpha)(z - \bar{\alpha}) = z^2 - 2\operatorname{Re}(\alpha)z + |\alpha|^2$ is a **real quadratic factor**.
3. Use polynomial division or compare coefficients to find the remaining factor.
4. Solve the remaining quadratic equation (completing the square or quadratic formula).

Example (2021F-Q6): $P(2 + 4i) = 0 \Rightarrow$ factor $z^2 - 4z + 20 \Rightarrow P = (z^2 - 4z + 20)(z^2 - 2z + 3) \Rightarrow z = 2 \pm 4i, 1 \pm \sqrt{2}i$.

Finding coefficients in reverse (2023F-Q2): given roots $1 + i, 2 + \sqrt{3}i \Rightarrow$ the conjugate roots are also roots $\Rightarrow Q(z) = (z^2 - 2z + 2)(z^2 - 4z + 7) \Rightarrow$ use the constant term 14 to deduce the real root z_0 , then expand and compare coefficients.

Past questions: 2016F-Q3, 2016S-F1, 2017F-Q2, 2018F-Q7(b)(c), 2019F-Q2, 2021F-Q6, 2022A-Q13(a)(b), 2023F-Q2, 2024F-Q7.

Pattern 4: Geometric meaning of multiplication (rotation + dilation)

Recognition: "Describe the geometric transformation performed by successive multiplication by z ".

Template (2 marks): "Multiplication by $w = r \operatorname{cis} \theta$ **dilates** the modulus by a factor r and **rotates** the vector **anticlockwise by θ** (or clockwise by $|\theta|$ if $\theta < 0$) about the origin." 1 mark for each of the two elements; for multiplication by z^{-1} , take the reciprocal of the scale factor and reverse the direction of rotation (2023A-Q10(d)).

Extensions

- Find all n for which zw^n lies in the second quadrant: each multiplication by w rotates by 30° , list the n for which $90^\circ < \arg w^n < 180^\circ$. When $|w| = 1$, it returns to the original position every 12 times, so the answer must be complete: $n = 3 + 12k, 4 + 12k, 5 + 12k$ (k a non-negative integer); writing only 3, 4, 5 does not earn the "all the possible values" mark (2025A-Q19(b)).
- Read the modulus and argument of z from the diagram, then write the polar form (2019A-Q12(b)).

Past questions: 2017A-Q10, 2019A-Q12, 2023A-Q10, 2025A-Q19.

Pattern 5: Loci — sketching from equations

Equation or inequality	Graph
$ z - a = r$	circle with centre a and radius r
$ z - a = z - b $	perpendicular bisector of the line segment joining a and b

$\text{Arg}(z - a) = \theta$	ray starting at a (endpoint a is not included in the locus; draw an open circle)
$ z - a + z - b = k$	ellipse when $k > a - b $; when $k = a - b $ it degenerates to the line segment ab (2022A-Q9(a))
$ z + 2 = z - i + \sqrt{5}$	the difference equals the distance between the two points $\Rightarrow$ a ray starting at i and extending in the direction $-2 \rightarrow i$; the endpoint i is included in the locus (solid dot), unlike an Arg-type ray (the 2020A-Q10(b) key has 1 mark specifically for this point)
$(z + i) \cdot \text{conj}(z + i) = 2$	$ z + i ^2 = 2 \Rightarrow$ centre $-i$, radius $\sqrt{2}$ (2022A-Q9(b))
$z - \bar{z} \leq 6i$	$2i \cdot \text{Im}(z) \leq 6i \Rightarrow \text{Im}(z) \leq 3$ (2024A-Q10(b))

Marking points for sketching: ✓ key points or centre ✓ boundary position ✓ solid/dashed boundary ✓ correct shaded region.

Past questions: 2016A-Q10(a)(b), 2016S-A18, 2020A-Q10(b), 2022A-Q9(a)(b), 2024A-Q10(b), 2025A-Q10(b).

Pattern 6: Loci — writing equations from diagrams

Approach: first recognise the shape, then write the equation using z (do not use x, y): circle $\Rightarrow |z - \text{centre}| \leq r$; ray $\Rightarrow \text{Arg}(z - \text{endpoint}) = \theta$; sector $\Rightarrow$ use two Arg inequalities plus one modulus inequality. The angle at the endpoint must be found exactly using trigonometry (2020A-Q10(a)).

Past questions: 2018A-Q11(a)(b), 2019A-Q10(a), 2020A-Q10(a), 2021A-Q11(d), 2022A-Q9(c), 2025A-Q10(a).

Pattern 7: Maximum and minimum values on loci

Approach: sketch $\rightarrow$ find the geometric position where the maximum/minimum occurs $\rightarrow$ calculate using Pythagoras' theorem or trigonometry.

- Maximum distance from a point on the circle to $c = |\text{centre} - c| + r$, minimum distance $= |\text{centre} - c| - r$ (when c is outside the circle; when c is inside the circle it is $r - |\text{centre} - c|$) (2016A-Q10(c), 2024A-Q10(a)(ii)).
- Maximum argument of a point on the circle: occurs at the **tangent** position. The θ found from $\sin \theta = r/d$ is only "the angle between the tangent and the line $O \rightarrow \text{centre}$ "; you must then add or subtract the argument of the centre, and note the argument range specified in the question (2018A-Q11(c): $0 \leq \arg z < 2\pi$, answer $2\pi - \alpha \approx 5.64$).
- Minimum distance from a point on the ray to i : draw a perpendicular (2019A-Q10(b)).

Marking points: ✓ indicate on the diagram the position where the maximum/minimum occurs ✓ write the expression ✓ exact value.

Pattern 8: Using vectors or geometry to find modulus and argument (appears in both CF and CA)

Approach: Draw $z + w$ as a parallelogram and $z - k$ as a triangle, then use:

- Equal moduli of two vectors $\Rightarrow$ rhombus $\Rightarrow$ the diagonal bisects the angle: $\arg(z + r) = \theta/2$ (2018F-Q3(b)(ii));
- Isosceles triangles, the inscribed angle theorem, the sine or cosine rule (2017F-Q6, 2021F-Q1(b), 2023A-Q12);
- Regular hexagon: when **centred at the origin**, adjacent vertices differ in argument by $\pi/3$ and all moduli are equal. In 2025F-Q7, vertex z_0 is **at the origin**, so the hexagon is not centred at the origin: the side vectors rotate by $\pi/3$ successively; using the cosine rule gives $|z_2| = |z_4| = \sqrt{3}r$ and $|z_3| = 2r$, so $|z_1 z_2 z_3 z_4 z_5| = r \cdot \sqrt{3}r \cdot 2r \cdot \sqrt{3}r \cdot r = 6r^5$; $z_3 = 2r \text{ cis}(\pi/3)$, $\text{Arg}(z_5 - z_3) = -5\pi/6$ (the 2025 report calls out the hexagon geometry in this question).

Past exam questions: 2017F-Q1, 2017F-Q6, 2018F-Q3, 2020A-Q11, 2020A-Q15, 2021F-Q1, 2023A-Q12, 2025F-Q7.

Pattern 9: Proof and technique questions

- Identities: use $\text{conj}(z) = r \text{cis}(-\theta)$, together with $\cos$ being even and $\sin$ being odd (2018A-Q10).
- Grouped summation: $n \cdot i^n$ is not itself periodic, but the sum of every 4 consecutive terms (terms $4m+1$ to $4m+4$) is always $2 - 2i$; there are 505 groups $\Rightarrow \sum_{n=1}^{2020} n i^n = 505 \times (2 - 2i) = 1010 - 1010i = 1010\sqrt{2} \text{cis}(-\pi/4)$ (2020F-Q8).
- Solving a real system of equations using complex numbers: $(a + bi)^3 = (a^3 - 3ab^2) + i(3a^2b - b^3) = 14 - 2\sqrt{5}i \Rightarrow$ taking the modulus: $(a^2 + b^2)^3 = 196 + 20 = 216 \Rightarrow a^2 + b^2 = 6$ (2023F-Q8).
- Use complex multiplication to prove the arctangent sum (2025A-Q19(c)).